THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

May 16, 2023 9:30 am – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the answer book.

Answer ALL EIGHT questions.

Notes:

- Write down your university number, seat number and model of calculator used on the cover of the answer book.
- The blank pages in the answer book can be used for rough work. You can also use them as additional answer space provided that you make a clear indication in the original answer space.
- Unless otherwise specified, you should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (12 marks) For this question, give answers only. No explanation is needed.

(a) Consider the double integral $\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2+y^2}\, dy\, dx$. If a change of variable is applied to transform the integral into one in polar coordinates, what will be the resulting integral? (Your answer should be a double integral involving r and θ. Do not evaluate the integral.)

(b) Find the determinant of the matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{bmatrix}$.

(c) Let $B = \begin{bmatrix} a & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{bmatrix}$ with $\det B = 3$. Find $\det \begin{bmatrix} a-2 & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{bmatrix}$.

(d) Evaluate the double integral $\iint_R x\, dA$, where R is the region bounded by the parabola $y = x^2 - 4$ and the straight line $y = 3x$.

(e) Using ONE iteration of Newton-Raphson's method, solve the system of equations

$$\begin{cases} x^2 + y^2 = 5 \\ x^3 + y^3 = 2 \end{cases}$$

with the initial guess $(2, -1)$. (Round your answer to 2 decimal places.)

(f) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & k \\ 3 & 1 & 1 \end{bmatrix}$ where k is a constant. If A is invertible, find all possible values of k.

2. (20 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like *is* and *are* do <u>not</u> indicate the singularity/plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. No explanation is needed.

EXAMPLE: Which of the following is correct?

(1) $1 + 2 = 3$.

(2) $1 + 1 = 2$.

(4) $1/0 = 1$.

(8) $1 - 1 = 0$. Answer: 11.

(a) Let f be a function in two variables x and y. If $f_x = 3y - x^2$ and $f_y = 3x - y$, which of the following must be a saddle point of f?

(1) $(0, 0)$

(2) $(0, 27)$

(4) $(9, 0)$

(8) $(9, 27)$

(b) Let $f(x, y)$ be a function defined for all real numbers x and y and whose partial derivatives of all orders exist and are continuous. If f has a local maximum at $(3, 4)$ and $f(3, 4) = 0$, which of the following must be positive?

(1) $f(2.9999, 3.9999)$

(2) $f_{xx}(3, 4)$

(4) $f_{yy}(3, 4)$

(8) $H_f(3, 4)$

(c) Which of the following can be transformed into a reduced row echelon matrix by one elementary row operation?

(1) $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(8) $\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(d) Which of the following makes sense only if the matrix A is square?

(1) The reduced row echelon of A

(2) The determinant of A

(4) The inverse of A

(8) The eigenvalues of A

(e) Let A be an $n \times n$ invertible matrix and λ is an eigenvalue of A. Which of the following is correct?

(1) $a\lambda$ is an eigenvalue of aA for any real number a.

(2) λ^n is an eigenvalue of A^n.

(4) λ^{-1} is an eigenvalue of A^{-1}.

(8) λ is an eigenvalue of A^T.

(f) If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

(1) $\text{rank}(A + B) = k$.

(2) $\text{rank}(A + B) \leq k$.

(4) $\text{rank}(A + B) \geq k$.

(8) $\text{rank}(A + B) = \text{rank}(A - B)$.

(g) Let $\mathbf{u}$ and $\mathbf{v}$ be vectors of $\mathbb{R}^n$, and let a be a scalar. Which of the following are identities?

(1) $\|\mathbf{u} \cdot \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|$.

(2) $\|a\mathbf{v}\| = a\|\mathbf{v}\|$.

(4) $\mathbf{v} \cdot \mathbf{v} = \sqrt{\|\mathbf{v}\|}$.

(8) $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$.

(h) Let $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ be linearly dependent vectors. Which of the following must be true?

(1) $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

(2) If $a\mathbf{x} + b\mathbf{y} + c\mathbf{z} = \mathbf{0}$, then $a \neq 0$.

(4) $\mathbf{x}$ is a linear combination of $\mathbf{y}$ and $\mathbf{z}$.

(8) $\text{Span}\{\mathbf{x}, \mathbf{y}\} = \text{Span}\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$.

(i) Which of the following are relative extreme points of $w = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$?

(1) $(0, 0)$

(2) $(0, 2)$

(4) $(1, 1)$

(8) $(-1, 1)$

(j) Let H be a solid hemisphere with radius a. Given that the density at each point of H is equal to its distance from the center of the base, which of the following integrals can be used to find the mass of H?

(1) $\displaystyle\int_0^{2\pi}\int_0^{a}\int_0^{\sqrt{a^2-r^2}} \sqrt{r^2+z^2}\, dz\, dr\, d\theta$

(2) $\displaystyle\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}\int_0^{\sqrt{a^2-x^2-y^2}} \sqrt{x^2+y^2+z^2}\, dz\, dy\, dx$

(4) $\displaystyle\int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho^3 \sin\phi\, d\rho\, d\phi\, d\theta$

(8) $\displaystyle\int_0^{2\pi}\int_0^{\pi/2}\int_0^{a} \rho^3 \sin\phi\, d\rho\, d\phi\, d\theta$

3. (18 marks) For each of the following statements, circle 'T' if it is true and 'F' if it is false, and then give a brief explanation.

(a) The vectors $\begin{bmatrix}8\\7\\6\end{bmatrix}$, $\begin{bmatrix}5\\4\\3\end{bmatrix}$ and $\begin{bmatrix}2\\1\\0\end{bmatrix}$ are linearly independent.

(b) Let A be a reduced row echelon matrix with rank 3. Then the third column of A must be $\mathbf{e}_3$.

(c) The function $f(x, y) = \begin{cases} \dfrac{x^3}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ continuous at $(0, 0)$.

(d) The linear system $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ can be solved by Cramer's rule.

(e) If a 2×3 matrix in reduced row echelon form has a zero row, it must have a zero column.

(f) The extreme values of a function defined on a bounded domain will always occur on the boundary of the domain.

4. (10 marks) Let $f(x, y) = xy(2 - x - y)$.

   (a) (6 marks) Find all critical points of f and determine their natures (relative maximum, relative minimum or saddle point).

   (b) (4 marks) Let T be the triangle (including the boundary and interior) with vertices $(0, 0)$, $(3, 0)$ and $(0, 3)$. Find the maximum and minimum values of f on T.

5. (10 marks) Find the maximum and minimum values of

$$f(x, y, z) = 3x^2 + y$$

subject to the constraints $4x - 3y = 9$ and $x^2 + z^2 = 9$.

6. (10 marks) Consider the surfaces

$$S_1 : \ r^2 + z^2 = 9 \quad \text{and} \quad S_2 : \ r = 3 \sin \theta$$

in cylindrical coordinates.

   (a) (2 marks) What are the shapes of S_1 and S_2? (Explanation is not needed.)

   (b) (8 marks) Find the volume of the solid that lies inside both S_1 and S_2.

7. (10 marks) Let $A = \begin{bmatrix} 5 & 0 & -1 \\ 4 & k & 2 \\ 3 & 0 & 1 \end{bmatrix}$ where k is a real number.

(a) (3 marks) Find the eigenvalues of A.

(b) (3 marks) Determine if A is diagonalizable when $k = -2$. If yes, diagonalize it (explanation is not needed); if not, state the reason.

(c) (4 marks) Find, with careful explanations, all values of k for which A is diagonalisable.

8. (10 marks) Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 6 \\ 7 \\ 8 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 9 \\ 10 \\ 11 \end{bmatrix}$.

(a) (2 marks) Without carrying out any numerical computations, explain why $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$ and $\mathbf{v}_4$ must be linearly dependent.

(b) (4 marks) Let $X = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $Y = \text{Span}\{\mathbf{v}_3, \mathbf{v}_4\}$. Is X equal to Y? Explain.

(c) (4 marks) Let $Z = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$. Is Z equal to $\mathbb{R}^3$? Explain.

******* **END OF PAPER** *******